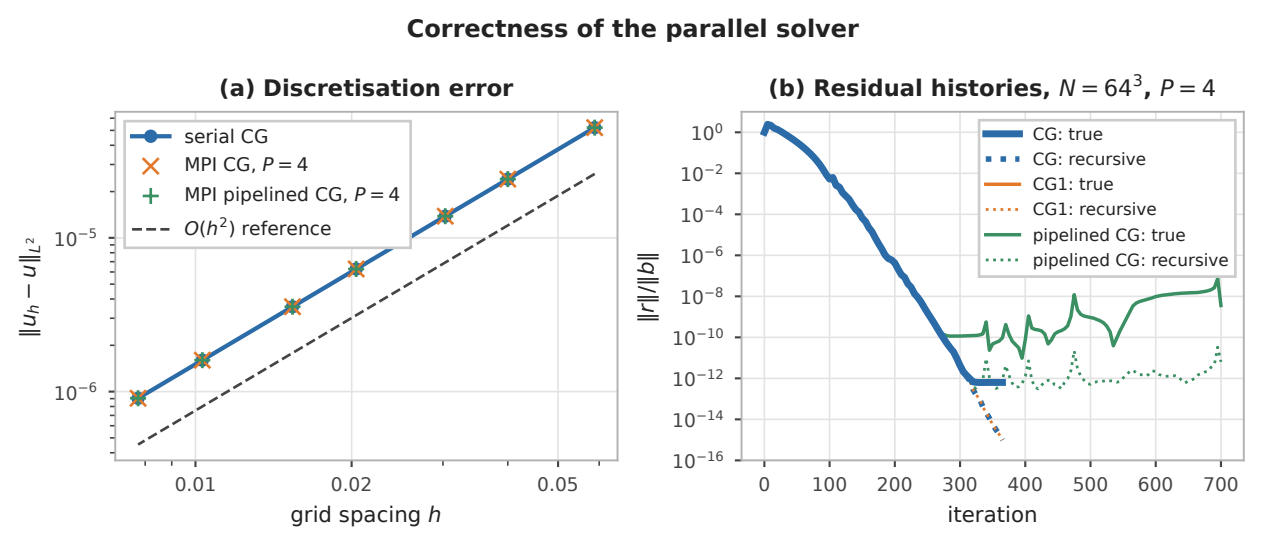
Correctness of the parallel solver
(a) Discretisation error
serial CG
MPI CG, $P = 4$
MPI pipelined CG, $P = 4$
$O(h^2)$ reference
$\|u_h - u\|_{L^2}$
$10^{-5}$
$10^{-6}$
0.01
0.02
0.05
grid spacing $h$
(b) Residual histories, $N = 64^3$, $P = 4$
CG: true
CG: recursive
CG1: true
CG1: recursive
pipelined CG: true
pipelined CG: recursive
$\|r\|/\|b\|$
$10^{0}$
$10^{-2}$
$10^{-4}$
$10^{-6}$
$10^{-8}$
$10^{-10}$
$10^{-12}$
$10^{-14}$
$10^{-16}$
0
100
200
300
400
500
600
700
iteration